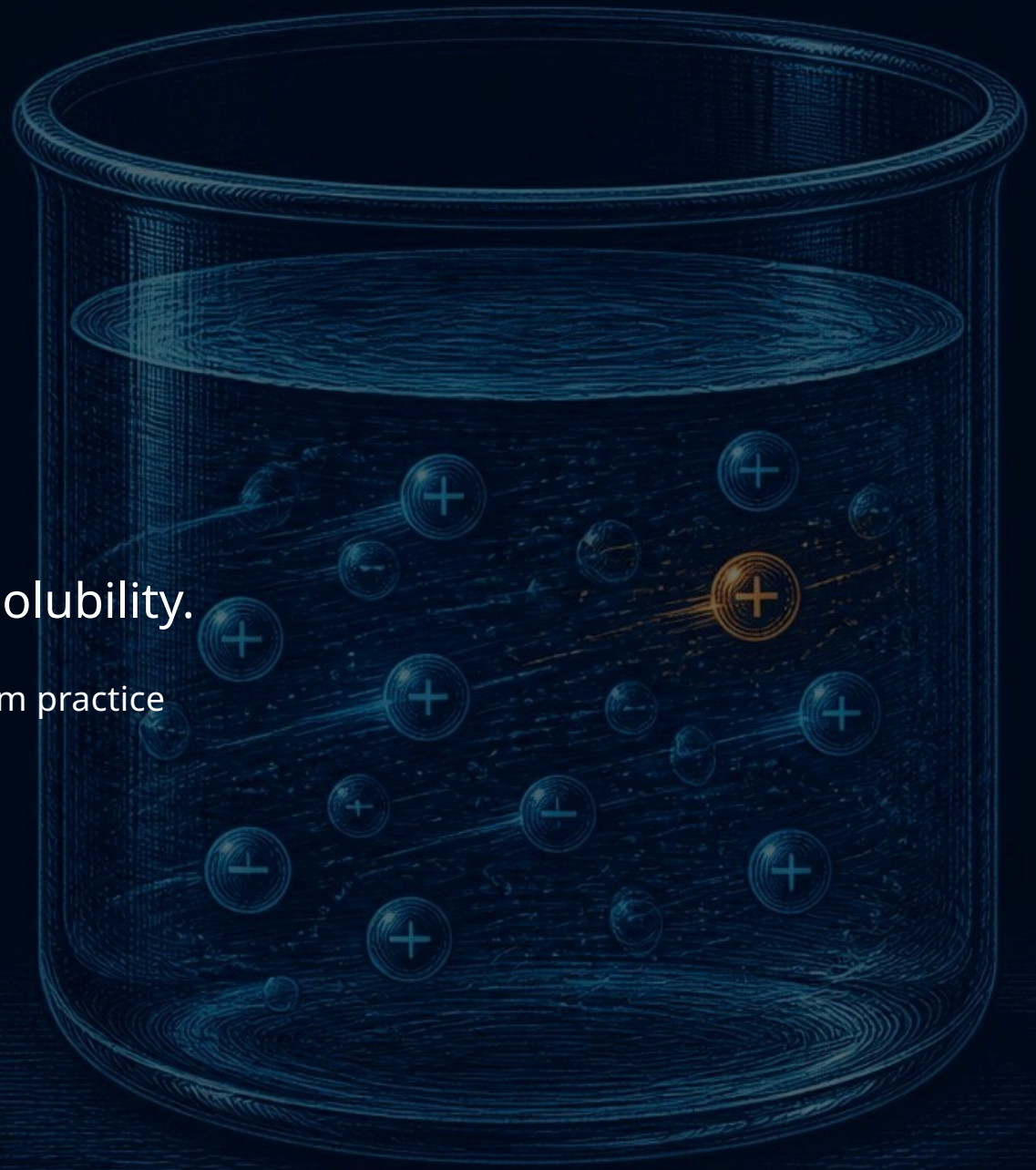
CLASS 12 CHEMISTRY · CHAPTER STUDY

Ionic Equilibrium

Ions, acids, bases, buffers, hydrolysis, and solubility.

Maharashtra State Board preparation · Board + entrance-exam practice



CHAPTER MAP

Choose an equilibrium concept

01

Electrolytes

Ions, ionisation, and degree of dissociation.

02

Acid–base theories

Arrhenius, Brønsted–Lowry, Lewis, and conjugate pairs.

03

Ionisation constants

Ka, Kb, pKa, pKb, and Ostwald's dilution law.

04

Water and pH

Kw, pH, pOH, and concentration scales.

05

Buffers

Common-ion effect and resistance to pH change.

06

Salts and solubility

Hydrolysis, Ksp, ionic product, and precipitation.

07

Practice path

Numericals, MCQs, board preparation, revision, and formula sheet.

EXAM PRIORITY · GUIDE

Board and entrance-exam focus by subtopic

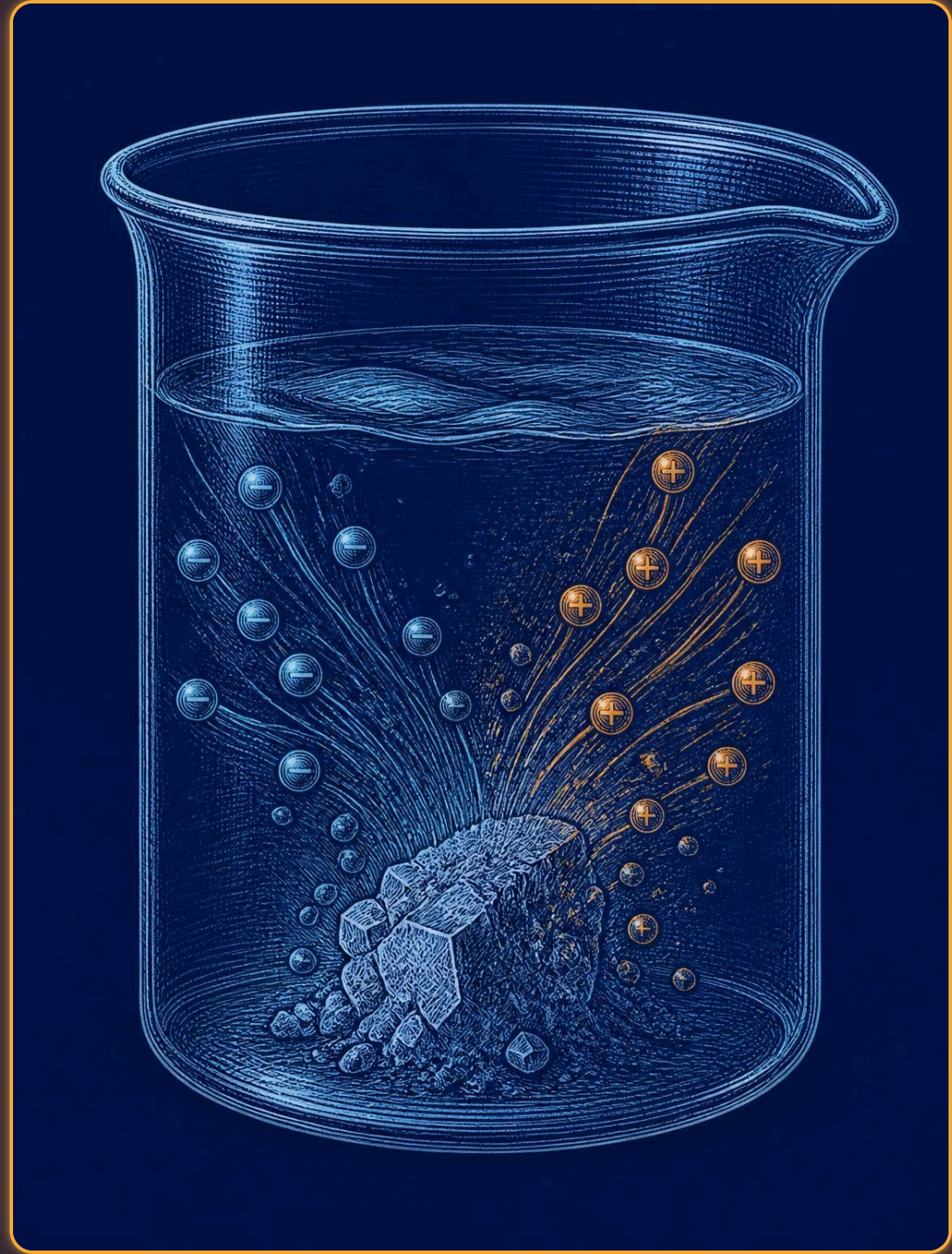
Focus	What to practise
Board	Definitions, equilibrium expressions, pH calculations, buffer action, hydrolysis, Ksp, and numerical reasoning.
Entrance	Rapid pH, Ka/Kb, degree of dissociation, common-ion, buffer, and precipitation calculations.
Use caution	Do not treat any chapter-wise entrance-exam mark count as fixed unless an official blueprint confirms it.

ORIENTATION

Electrolytes form ions in solution

Ionisation extent classifies electrolytes; aqueous ions conduct electricity.

Strong electrolytes ionise almost completely. Weak electrolytes ionise partially and establish dynamic equilibrium. Non-electrolytes do not produce ions in aqueous solution.



MEASURE THE EXTENT

Degree of dissociation measures ionisation

Definition

α = ionised amount / initial amount

For a weak electrolyte, dilution usually increases α.

Keep terms separate

α describes the fraction ionised.

Concentration describes amount per volume.

Conductivity describes charge transport.

PRACTICE CHECK · UNSOURCED

Check yourself: electrolytes

MCQ 1. Classify a substance as strong, weak, or non-electrolyte.

Use the extent of ion formation in aqueous solution.

▼ Show answer

> **Answer.** Strong: almost complete ionisation. Weak: partial ionisation. Non-electrolyte: no appreciable ions.

MCQ 2. Predict how dilution changes the degree of dissociation.

Think about the equilibrium of a weak electrolyte.

▼ Show answer

> **Answer.** Dilution usually increases α for a weak electrolyte.

Practice questions, not verified past-paper questions.

THREE LENSES

Acid–base theories give different lenses

Each theory expands the range of reactions that can be described.

Arrhenius

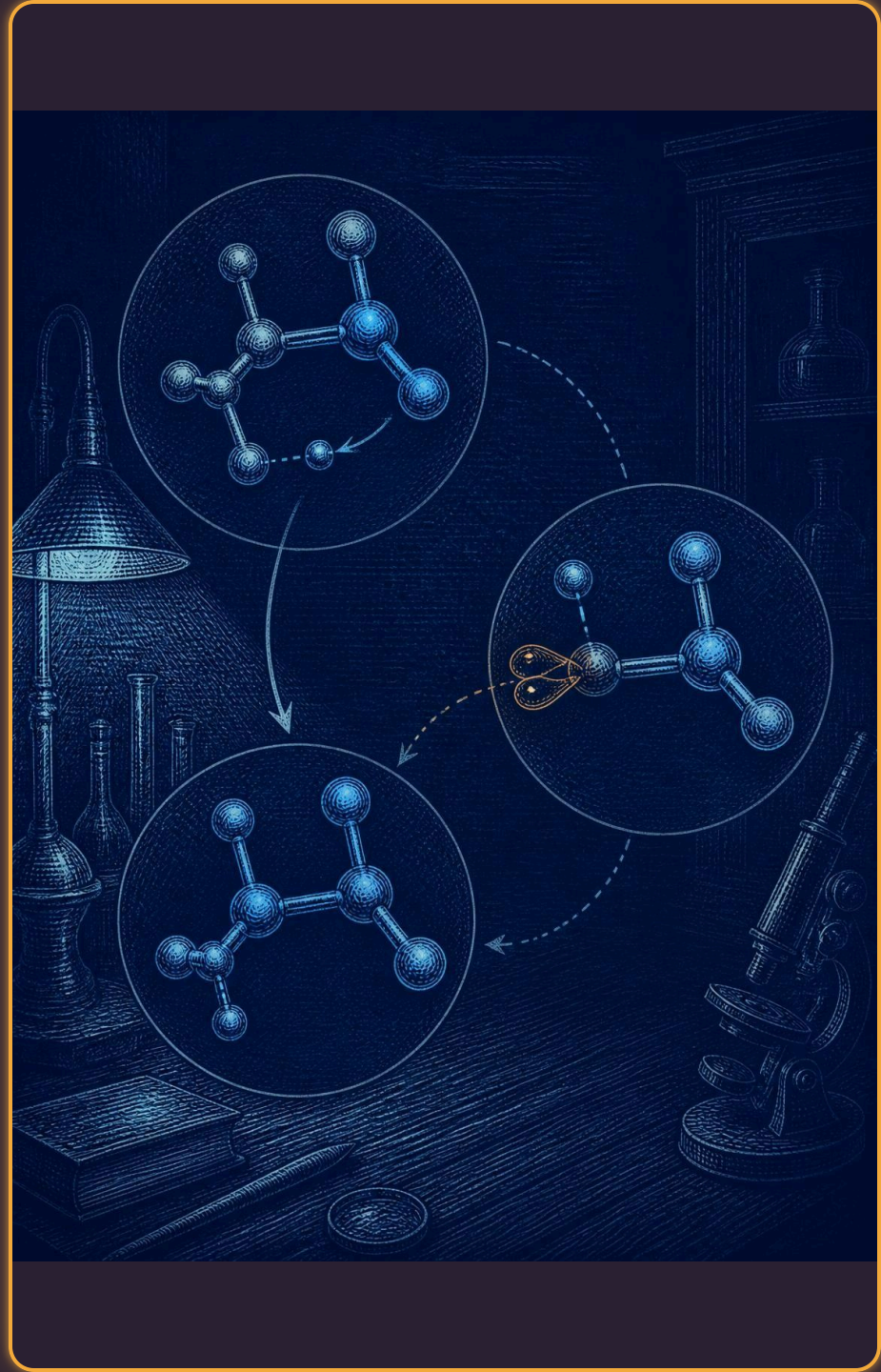
Acid produces H^+ in water; base produces OH^-.

Brønsted–Lowry

Acid donates a proton; base accepts a proton.

Lewis

Acid accepts an electron pair; base donates one.



PROTON TRANSFER

Conjugate acid–base pairs

An acid and its conjugate base differ by one proton. A stronger acid has a weaker conjugate base.

Track the proton, not just the labels.



PRACTICE CHECK · UNSOURCED

Check yourself: acid–base theories

MCQ 3. Identify the Brønsted acid and base in a reaction.

Find the proton donor and proton acceptor.

▼ Show answer

> **Answer.** The Brønsted acid donates H^+; the Brønsted base accepts H^+.

MCQ 4. Identify the Lewis acid and Lewis base in a coordination reaction.

Look for electron-pair acceptance and donation.

▼ Show answer

> **Answer.** The Lewis acid accepts an electron pair; the Lewis base donates it.

Practice questions, not verified past-paper questions.

WEAK ELECTROLYTES

Ionisation constants measure weak-acid strength

For **HA ⇌ H⁺ + A⁻**:

$$Ka = [H^+][A^-]/[HA]$$

For a weak base, Kb uses the corresponding equilibrium concentrations.

Read the logarithm

pKa = −log Ka

Smaller pKa means a stronger acid.

DILUTION RELATION

Ostwald's dilution law

For a weak electrolyte of concentration C and degree of dissociation α:

$$K = C\alpha^2/(1-\alpha)$$

For small α, **$K \approx C\alpha^2$**.

Interpretation

Dilution increases α.

The equilibrium constant does not change at fixed temperature.

The composition shifts; K remains characteristic of the equilibrium.

PRACTICE · WORKED EXAMPLE 1

Weak-acid dissociation

Given

Weak-acid concentration C and Ka.

Find

α and approximate ion concentrations.

Formula

$Ka \approx C\alpha^2$

Substitution

Insert C and Ka with units.

Calculation

$\alpha \approx \sqrt{(Ka/C)}$

Answer

Report α and $[H^+] \approx C\alpha$.

Practice method, not a verified past-paper question. Check that α is small before using the approximation.

PRACTICE CHECK · UNSOURCED

Check yourself: Ka and Kb

MCQ 5. Compare acid strength from Ka or pKa values.

A larger Ka means stronger acid; a smaller pKa means stronger acid.

▼ Show answer

> **Answer.** Choose the larger Ka, or the smaller pKa.

MCQ 6. Use Ka and concentration to estimate $[H^+]$ for a weak acid.

Use the small-dissociation approximation only when justified.

▼ Show answer

> **Answer.** $[H^+] \approx \sqrt{(K_a C)}$.

Practice questions, not verified past-paper questions.

AUTOIONISATION OF WATER

Water ionises to a very small extent

Simplified equilibrium:

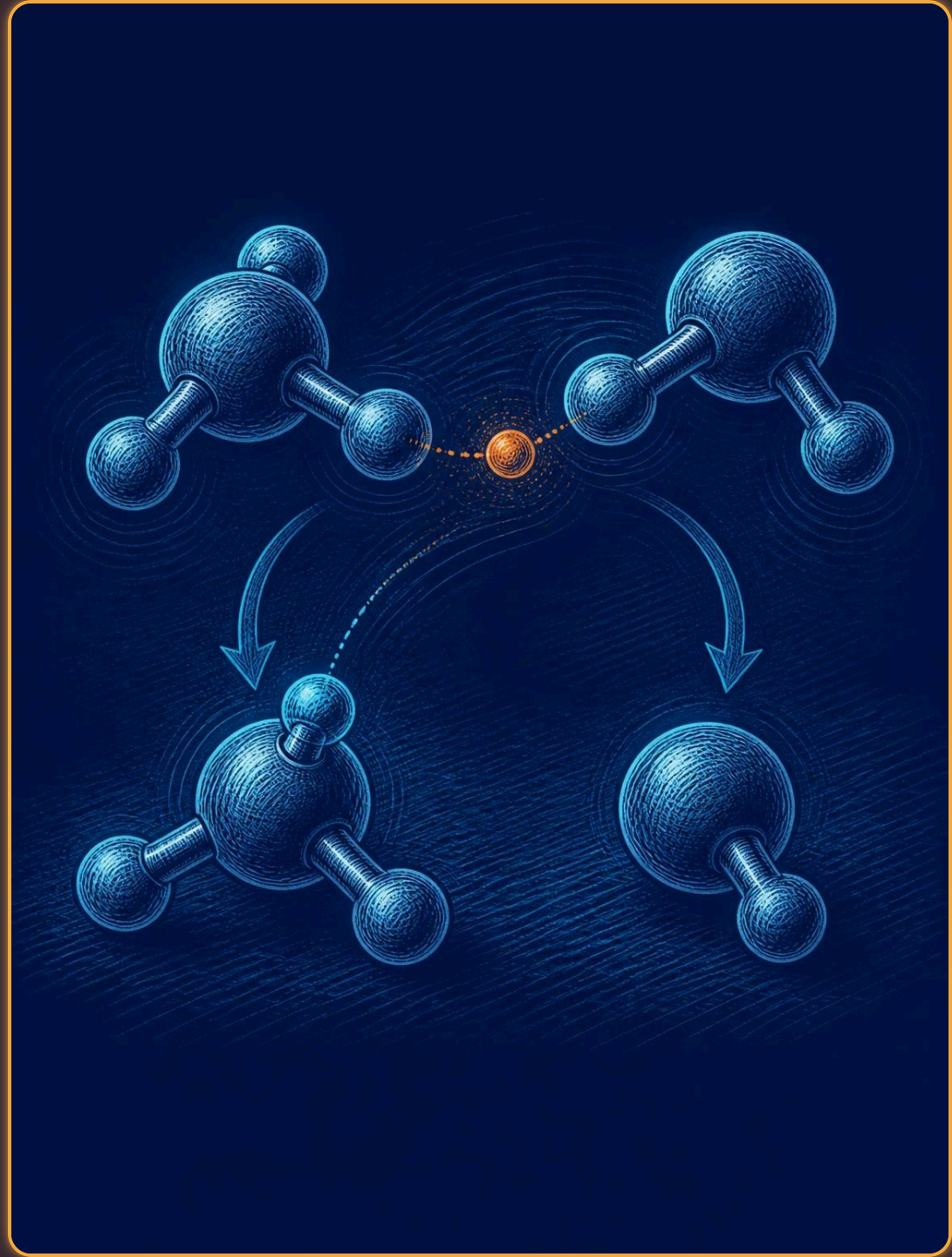
$$H_2O \rightleftharpoons H^+ + OH^-$$

More accurately, hydronium forms:

$$2H_2O \rightleftharpoons H_3O^+ + OH^-$$

Kw = $[H^+][OH^-]$

At 25 °C, neutral water has equal H^+ and OH^- concentrations.



LOGARITHMIC SCALE

pH and pOH compress concentration scales

$$pH = -\log[H^+]$$

$$pOH = -\log[OH^-]$$

Acidic solutions have larger $[H^+]$.

Basic solutions have larger $[OH^-]$.

At 25 °C: pH + pOH = 14.

At 25 °C, pH below 7 is acidic; pH above 7 is basic.

PRACTICE · WORKED EXAMPLE 2

pH of a strong acid

Given

Molar concentration C of a strong monoprotic acid.

Find

$[H^+]$, pH, and pOH.

Formula

$[H^+] = C$

Substitution

Use C in mol L^{-1}.

Calculation

$pH = -\log C$; then $pOH = 14 - pH$.

Answer

Report $[H^+]$, pH, and pOH.

Practice method, not a verified past-paper question. At 25 °C, use pH + pOH = 14.

PRACTICE CHECK · UNSOURCED

Check yourself: water and pH

MCQ 7. Find pH from $[H^+] = 1 \times 10^{-3}$ mol L^{-1}.

▼ Show answer

> **Answer: pH = 3.** Use $pH = -\log[H^+]$.

MCQ 8. Find $[OH^-]$ from pH at 25 °C.

▼ Show answer

> **Answer.** First find $pOH = 14 - pH$, then $[OH^-] = 10^{-pOH}$ mol L^{-1}.

Practice questions, not verified past-paper questions.

APPROXIMATION TOOL

Weak-acid and weak-base pH

Weak acid

$$[H^+] \approx \sqrt{(KaC)}$$

Then calculate pH from [H⁺].

Weak base

$$[OH^-] \approx \sqrt{(KbC)}$$

Then calculate pOH and pH.

Check that the estimated degree of dissociation is small before using either approximation.

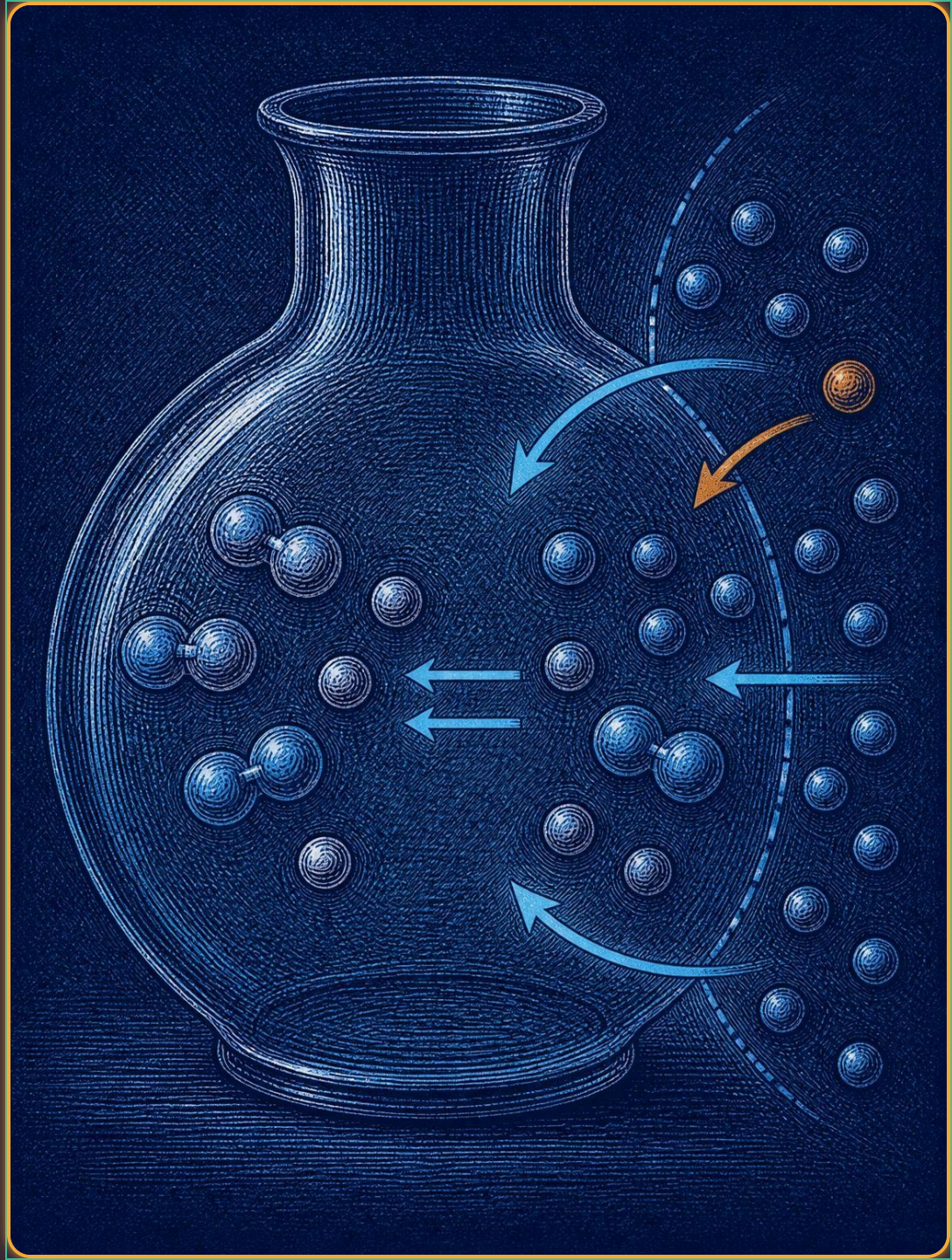
EQUILIBRIUM SHIFT

Common-ion effect shifts equilibrium

Adding an ion already present suppresses dissociation of a weak electrolyte.

Use Le Châtelier's principle: the equilibrium shifts away from the added ion.

Applications include buffers, selective precipitation, and qualitative analysis.



CONTROLLED PH

Buffer solutions resist pH change

Acidic buffer

Weak acid plus its conjugate salt.

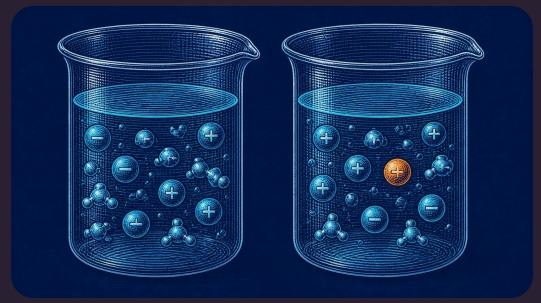
pH = pKa + log([salt]/[acid])

Basic buffer

Weak base plus its conjugate salt.

Use the corresponding Kb relationship.

The conjugate pair consumes small added amounts of acid or base, limiting the pH change.



PRACTICE · WORKED EXAMPLE 3

Acidic buffer

Given

pKa and the salt-to-acid concentration ratio.

Find

pH of the buffer.

Formula

pH = pKa + log([salt]/[acid])

Substitution

Insert the ratio without changing its order.

Calculation

Evaluate the logarithm and add pKa.

Answer

Report pH with suitable precision.

Practice method, not a verified past-paper question. The ratio is salt over acid.

PRACTICE CHECK · UNSOURCED

Check yourself: buffers

MCQ 9. Which mixture forms an acidic buffer?

Look for a weak acid and its conjugate salt.

▼ Show answer

> **Answer.** A weak acid plus the salt of its conjugate base forms an acidic buffer.

MCQ 10. Predict the pH change when a small amount of acid is added to a buffer.

Compare the change with the same acid added to pure water.

▼ Show answer

> **Answer.** The pH decreases only slightly because the buffer consumes much of the added H^+.

Practice questions, not verified past-paper questions.

SALT SOLUTIONS

Salt hydrolysis changes solution pH

Strong acid + strong base

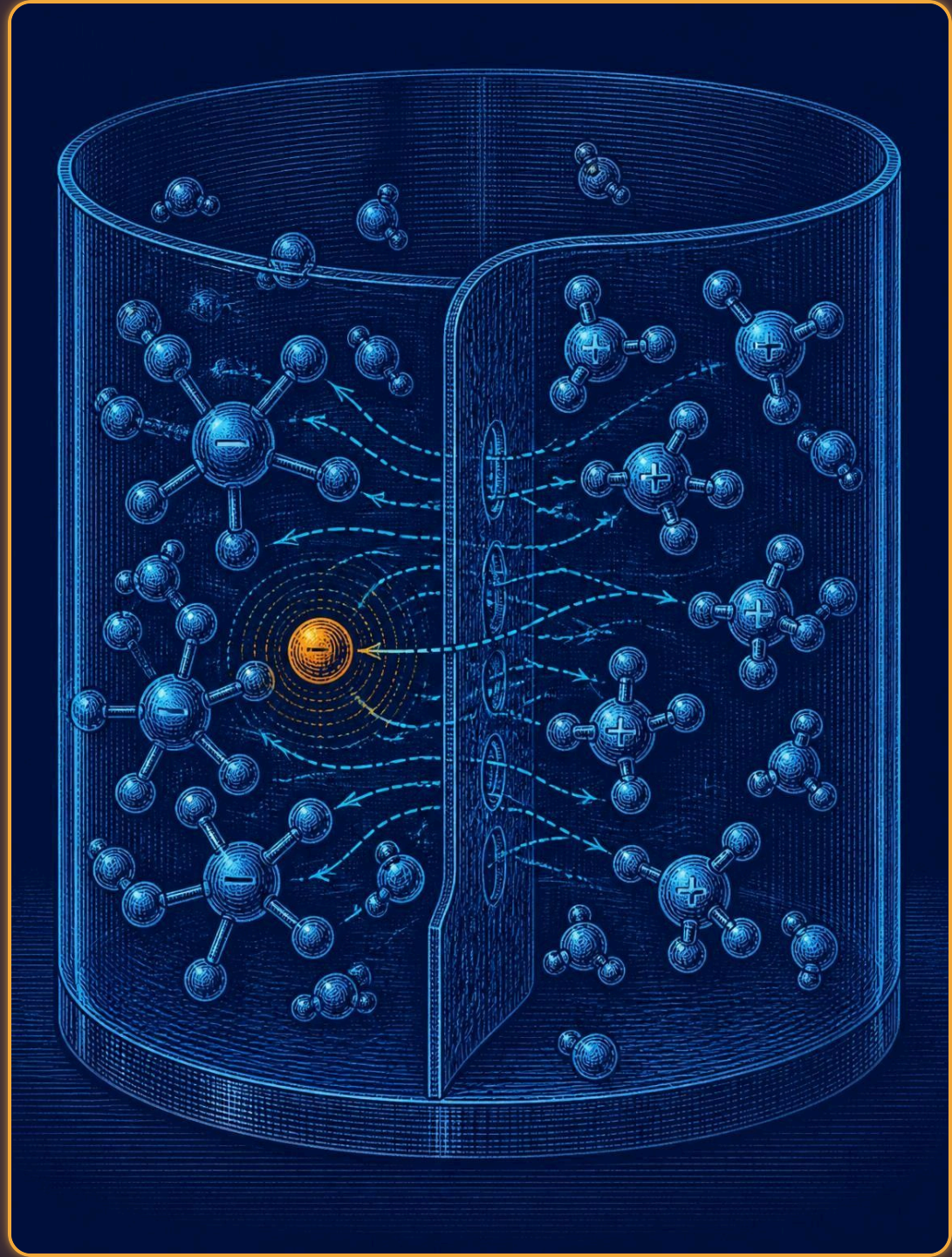
Approximately neutral.

Weak acid + strong base

Basic solution.

Strong acid + weak base

Acidic solution.



HYDROLYSIS RELATIONS

Hydrolysis constants and degree of hydrolysis

Connect constants

Relate Kh to Ka, Kb, and Kw for the salt type.

Estimate extent

Use concentration and Kh to estimate hydrolysis when the approximation is appropriate.

Check the model

Separate hydrolysis from simple dilution.

Report clearly

State whether the final solution is acidic, basic, or approximately neutral.

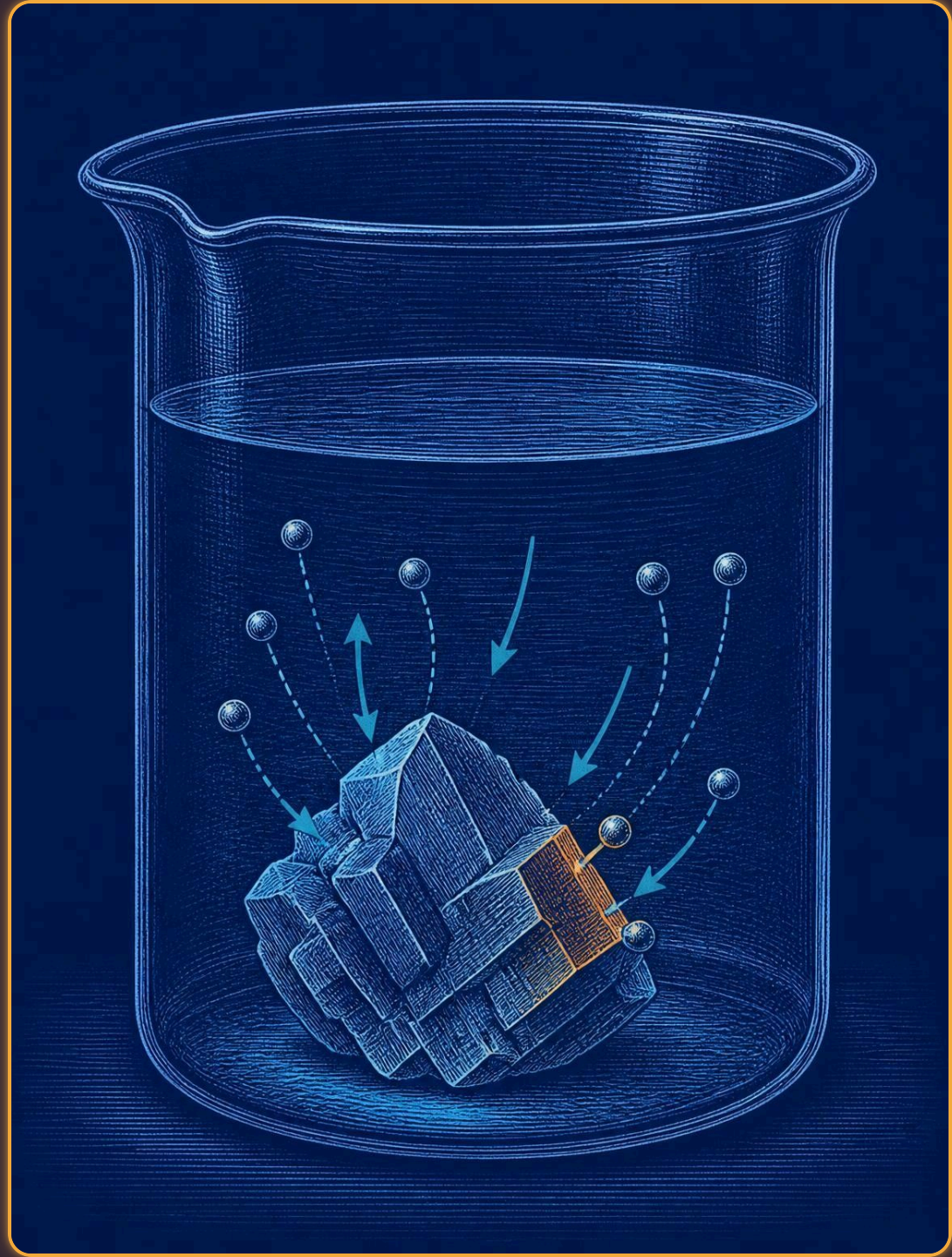
SPARINGLY SOLUBLE SALTS

Solubility equilibrium and Ksp

A sparingly soluble salt establishes equilibrium between the solid and its ions.

Ksp is the product of equilibrium ion concentrations, each raised to its stoichiometric power.

Use molar solubility s to build the expression.



PRACTICE · WORKED EXAMPLE 4

Ksp and precipitation

Given

Ksp and the salt stoichiometry.

Find

Molar solubility or precipitation tendency.

Formula

Build Ksp from the dissolution equation.

Substitution

Write each ion concentration in terms of s.

Calculation

Compare ionic product Q with Ksp.

Answer

Q > Ksp: precipitation is predicted.

Practice method, not a verified past-paper question. Keep stoichiometric powers visible.

PRACTICE CHECK · UNSOURCED

Check yourself: Ksp

MCQ 11. Write the Ksp expression for a 1:2 salt.

For $MX_2(s) \rightleftharpoons M^{2+} + 2X^-$, include the squared anion concentration.

▼ Show answer

> **Answer.** Ksp = $[M^{2+}][X^-]^2$.

MCQ 12. Predict precipitation by comparing ionic product and Ksp.

▼ Show answer

> **Answer.** Q > Ksp: precipitation; Q = Ksp: saturation; Q < Ksp: no precipitation predicted.

Practice questions, not verified past-paper questions.

QUICK REVISION

Formula sheet: Ionic Equilibrium

Topic	Formula
Weak acid/base	**$K_a = [H^+][A^-]/[HA]$; $K_b = [BH^+][OH^-]/[B]$**
Water and pH	**$K_w = [H^+][OH^-]$; $pH = -\log[H^+]$; $pOH = -\log[OH^-]$**
At 25 °C	**$pH + pOH = 14$**
Approximation	**$[H^+] \approx \sqrt{(K_aC)}$; $[OH^-] \approx \sqrt{(K_bC)}$**
Buffer	**$pH = pK_a + \log([salt]/[acid])$**
Solubility	**Ksp** is the ionic concentration product at saturation.

ERROR CHECK

Revision notes and common mistakes

Equilibrium values

Use equilibrium concentrations, not initial concentrations, in Ka, Kb, and Ksp.

Notation

Do not confuse pKa with Ka or pH with $[H^+]$.

Stoichiometry

Check stoichiometric powers in Ksp expressions.

Temperature

Use 25 °C only when applying pH + pOH = 14.

Buffer identity

Confirm the acid/base pair before choosing a buffer formula.

EXAM PREPARATION

Board exam preparation

01

Define

Electrolytes, acids, bases, equilibrium constants, buffers, hydrolysis, and Ksp.

Explain

Ionisation, acid–base theories, conjugate pairs, common-ion effect, buffer action, and precipitation conditions.

Derive

Ostwald's dilution law, pH derivations and relations, and hydrolysis expressions.

02

Calculate

Ka/Kb, pH, buffer, hydrolysis, solubility, and ionic-product numericals.

Check

Finish with units, logarithms, approximations, and stoichiometric powers.

All questions in this deck are practice unless a verified board-paper source is identified.

LAST-MINUTE CHECK

Last-minute revision checklist

+=x

Write the equilibrium equation before the constant expression.

Track stoichiometric powers and units.

Use logarithms carefully and keep concentrations in mol L^{-1}.

Check whether the solution is strong, weak, buffered, hydrolysed, or saturated.

Finish with one pH, one buffer, one hydrolysis, and one Ksp problem.

REFERENCES AND SOURCE NOTE

References and source note

Use the current Maharashtra State Board / Balbharati Standard XII Chemistry textbook for final terminology and scope.

Do not claim previous-year status for unsourced questions.

Optional videos, if added later, are supplementary and not official.

Check final exam wording and prescribed scope against your current textbook.

